

Birhane Hiwot Sunday School — QA Test Case Document

(ብርሃነ አይወት ሰንበት ትምህርት ቤት የፈተና እና የጥራት ቁጥጥር ሰነድ)

Application: Birhane Hiwot Sunday School Attendance & Education Management System (Web & API)
Version: 1.0.0
Last updated: August 26, 2026
Related assets:

- [Postman Collection](file:///c:/Users/Ym/Desktop/b/ss/Birhane-Hiwot-sunday-school-attendance/docs/qa/Birhane-Hiwot-API.postman_collection.json)
- [Postman Environment](file:///c:/Users/Ym/Desktop/b/ss/Birhane-Hiwot-sunday-school-attendance/docs/qa/Birhane-Hiwot.postman_environment.json)
- [OpenAPI Specification](file:///c:/Users/Ym/Desktop/b/ss/Birhane-Hiwot-sunday-school-attendance/public/openapi.json) (Interactive Swagger UI at [/api-docs](#))

1. Test Scope

In scope	Out of scope
Authentication & session management (NextAuth Credentials & Mobile JWT)	Third-party Google Sheets cloud availability (tested via health endpoint)
Role-based access control (Super Admin, HR Admin, Education Admin, Facilitators, Teachers)	Network hardware benchmarking
Student CRUD, check duplicate, grade counts, QR codes, registration gating	Third-party SMS/telecom gateways
Bulk attendance marking with distributed locking, filtering & summaries	Penetration testing of third-party hosting infrastructure
Attendance Facilitator & Education Facilitator user management	
Teacher profiles, curriculum subjects grouping, batch operations, & assignments	
Student academic results, score aggregation, & university letter grade mapping	
Student registration request submission & admin approval workflow	
13-month Ethiopian fee payment tracking	

In scope	Out of scope
HMAC-SHA256 signed QR code generation & verification	
Excel and PDF reporting & export	

2. Test Environment

Item	Value
Base URL (Local)	http://localhost:3000
Base URL (Production / Staging)	https://birhane-hiwot-sunday-school-attenda.vercel.app
Database	MongoDB Atlas (MONGODB_URI)
Web Auth	NextAuth Credentials provider, JWT session (NEXTAUTH_SECRET), 24h expiration
Mobile Auth	Direct signed HS256 JWT (/api/mobile/auth), 7-day expiration
Distributed Locks	Redis (REDIS_URL) / In-memory fallback for date & Unique_ID concurrency
Calendar System	Ethiopian Calendar (13 Ethiopian months, Academic Year 2018, 2017)

Authentic Test Accounts

Role	Canonical Test Name	Purpose / Responsibility
Super Admin	ለቀ ትጉሃን ገብረመስቀል	Full system administration; manages department admins via /api/admin-users
HR Admin	ቀሲስ ዳዊት በቀለ	HR module, attendance facilitators, student records, permission toggles
Education Admin	ለቀ ዲያቆናት ሰላሞን	Education module, teachers, subjects, education facilitators, academic results
Attendance Facilitator	ዲ/ን ቴዎድሮስ ካሳሁን	Marks attendance for assigned grades; creates students when canAddStudent is true
Education Facilitator	መርጌታ ዘርዓይ በርሄ	Educational result entry, exam recording, grade reports
Teacher	መምህረ አስፋው ወልደሥላሴ	Assigned classes, subjects, and student roster access

3. Test Case Summary

Module	Total Cases	Priority High
Authentication & Setup (Web & Mobile)	18	14
Authorization / RBAC	18	16
Student Management	31	22
Attendance	16	13
Facilitators & Education Facilitators	15	12
Teachers, Subjects & Assignments	17	11
Category Periods (Registration Windows)	5	3
Student Results	13	9
Payments (13 Ethiopian Months)	8	6
Student Requests Approval Workflow	10	8
QR Code Generation & Verification	6	5
Reports & Export	8	4
Negative & Security Edge Cases	8	4
Total	173	127

4. Authentication & Setup

ID	Test Case	Steps	Expected Result	Priority
AUTH-001	Valid web login	Enter valid email (superadmin@birhanehiwot.org) and password on /login, submit	Redirect to role dashboard; next-auth.session-token cookie set	High
AUTH-002	Invalid password	Enter valid email, wrong password	401 error message; user remains on login page	High
AUTH-003	Invalid email	Enter non-existent email address	401 error message; no session created	High
AUTH-004	Empty fields	Submit with blank email or password	Form validation prevents submit	High
AUTH-005	Case-insensitive email	Login with SUPERADMIN@birhanehiwot.org	Login succeeds seamlessly	Medium
AUTH-006	Session persistence	Login, refresh browser page	User remains authenticated via JWT session	High
AUTH-007	Session expiry	Invalidate cookie or simulate 24h expiration	User redirected to /login	Medium

ID	Test Case	Steps	Expected Result	Priority
AUTH-008	Web logout	Click logout or POST <code>/api/auth/signout</code>	Session cookie cleared; redirected to login	High
AUTH-009	Protected route without auth	Navigate to <code>/admin/dashboard</code> while logged out	Redirected to <code>/login</code> with <code>callbackUrl</code>	High
AUTH-010	Web session API check	GET <code>/api/auth/session</code> while authenticated	Returns <code>{ user: { id, name, email, role, grade, canAddStudent } }</code>	Medium
AUTH-011	CSRF token flow	GET <code>/api/auth/csrf</code> then POST to credentials callback	Login succeeds with valid CSRF token	Medium
AUTH-012	Super Admin setup (one-time)	POST <code>/api/setup/super-admin</code> with <code>confirm: "CREATE_SUPER_ADMIN"</code> , email, and password	201 Created, <code>{ message, created: true, userId }</code>	High
AUTH-013	Setup — invalid confirm token	POST setup with wrong <code>confirm</code> string	400 <code>Setup confirmation is invalid</code>	High
AUTH-014	Setup — password too short	POST setup with password < 6 characters	400 <code>Password must be at least 6 characters</code>	Medium
AUTH-015	Setup — idempotent when exists	POST setup again after Super Admin exists	200 OK, <code>{ created: false, message: "Super Admin already exists" }</code>	Medium
MOB-001	Mobile login valid credentials	POST <code>/api/mobile/auth</code> with valid facilitator credentials	200 OK, returns <code>{ token, user: { id, email, name, role, grade } }</code>	High
MOB-002	Mobile login invalid credentials	POST <code>/api/mobile/auth</code> with wrong password	401 <code>Invalid email or password</code>	High
MOB-003	Mobile CORS preflight	OPTIONS <code>/api/mobile/auth</code>	204 No Content with <code>Access-Control-Allow-Origin: *</code>	Medium

5. Authorization (Role-Based Access)

ID	Test Case	Role	Route / Action	Expected Result	Priority
RBAC-001	Super Admin full access	Super Admin	/super-admin/* pages	Access granted	High
RBAC-002	Non-super blocked from super-admin	HR Admin	/super-admin/dashboard	Access denied / redirect	High
RBAC-003	HR Admin HR module	HR Admin	/hr/*	Access granted	High
RBAC-004	HR Admin blocked from education	HR Admin	/education/*	Access denied	High
RBAC-005	Education Admin education module	Education Admin	/education/*	Access granted	High
RBAC-006	Attendance Facilitator attendance	Attendance Facilitator	/facilitator/attendance/* (assigned grades only)	Access granted	High
RBAC-007	Attendance Facilitator results blocked	Attendance Facilitator	/facilitator/results/*	Access denied	High
RBAC-008	Education Facilitator results	Education Facilitator	/facilitator/results/*	Access granted	High
RBAC-009	HR Admin student records access	HR Admin	/admin/students , /admin/facilitators , /admin/reports	Access granted	High
RBAC-010	HR Admin blocked from unauthorized routes	HR Admin	/admin/dashboard (if restricted)	Access denied	Medium
RBAC-011	Education Admin facilitator & report access	Education Admin	/admin/facilitators , /admin/reports	Access granted	High
RBAC-012	Facilitators API — list scoped by role	Attendance Facilitator	GET /api/facilitators	200 OK with only Attendance Facilitator records	High

ID	Test Case	Role	Route / Action	Expected Result	Priority
RBAC-013	Facilitators API — unauthenticated	No session	GET <code>/api/facilitators</code>	403 Forbidden	High
RBAC-014	Teachers API — manage restricted	Education Admin or Super Admin	POST <code>/api/teachers</code>	201 Created	High
RBAC-015	Teachers API — forbidden for other roles	Attendance Facilitator	POST <code>/api/teachers</code>	403 Forbidden	High
RBAC-016	Admin users API — Super Admin only	HR Admin	GET <code>/api/admin-users</code>	403 Forbidden	High
RBAC-017	Admin users API — Super Admin allowed	Super Admin	GET <code>/api/admin-users</code>	200 OK (lists HR Admin & Education Admin accounts)	High
RBAC-018	Dynamic permission refresh without re-login	HR Admin	PATCH <code>/api/facilitators/{id} {canAddStudent: true }</code>	Facilitator can add students on next action without re-logging in	High

6. Student Management

ID	Test Case	Steps	Expected Result	Priority
STU-001	List all students	GET <code>/api/students</code>	200 OK, array of students (with <code>photo_data_url</code> and <code>qr_code</code> excluded)	High
STU-002	Filter by grade(s)	GET <code>/api/students?grade=አንደኛ ክፍል</code> (repeatable param)	Returns students belonging to specified grade(s)	High
STU-003	Filter by academic year & sex	GET <code>/api/students?academicYear=2018&sex=Male</code>	Returns filtered male students in 2018	High

ID	Test Case	Steps	Expected Result	Priority
STU-004	Get by Unique ID (query)	GET <code>/api/students?uniqueId=BH-2018-001</code>	200 OK single student document, or 404 if missing	High
STU-005	Get by Mongo ID or Unique ID (path)	GET <code>/api/students/BH-2018-001</code>	200 OK full student record	High
STU-006	Create student — Super Admin	POST <code>/api/students</code> with authentic Amharic data (<code>First_Name: "አማኑኤል"</code> , <code>Father_Name: "ተክሌ"</code> , <code>Grade: "አንደኛ ክፍል"</code> , <code>Academic_Year: "2018"</code> , <code>userRole: "Super Admin"</code>)	201 Created, <code>{ _id }</code> returned; signed QR generated	High
STU-007	Create student — missing required field	POST omitting one of <code>Unique_ID</code> , <code>First_Name</code> , <code>Father_Name</code> , <code>Academic_Year</code> , <code>Grade</code>	400 Bad Request <code>{ error: "<field> is required" }</code>	High
STU-008	Concurrency lock on Unique_ID	POST two student creations with same <code>Unique_ID: "BH-2018-001"</code> simultaneously	Distributed lock prevents race condition; second request fails gracefully	High
STU-009	Facilitator without permission	POST student as Attendance Facilitator with <code>canAddStudent: false</code>	403 Forbidden <code>{ code: "ADD_STUDENT_DENIED" }</code>	High
STU-010	Facilitator with permission	Enable <code>canAddStudent</code> ; facilitator POSTs student in assigned grade	201 Created	High
STU-011	Facilitator wrong grade	Facilitator assigned to <code>አንደኛ ክፍል</code> attempts to create student in <code>ስድስተኛ ክፍል</code>	403 Forbidden <code>{ code: "RESTRICTED_GRADE" }</code>	High
STU-012	Facilitator missing userEmail	POST as Attendance Facilitator without <code>userEmail</code>	403 Forbidden "User email is required..."	Medium
STU-013	Non-admin student creation blocked	POST new student with <code>userRole: "Teacher"</code>	403 Forbidden "You do not have permission to add students."	High
STU-014	Invalid photo format	POST with non-JPEG/PNG <code>photo_data_url</code>	400 "photo_data_url must be a JPG or PNG data URL"	Medium

ID	Test Case	Steps	Expected Result	Priority
STU-015	Update student (PUT)	PUT <code>/api/students/{id}</code> with updated fields (<code>Phone_Number: "0911998877"</code>)	200 OK, updated student returned	High
STU-016	Regenerate QR on Unique_ID update	PUT changing <code>Unique_ID</code> from <code>BH-2018-001</code> to <code>BH-2018-002</code>	New signed <code>qr_code</code> generated automatically	Medium
STU-017	Update student photo (PATCH)	PATCH <code>/api/students/{id}</code> with valid PNG base64 data URL	200 OK, photo updated	Medium
STU-018	Regenerate QR code (PATCH)	PATCH <code>/api/students/{id}</code> with <code>{ "generateQR": true }</code>	200 OK, 300px PNG QR data URL regenerated	Medium
STU-019	PATCH with empty/invalid payload	PATCH <code>/api/students/{id}</code> with <code>{}</code>	400 <code>No valid fields to update</code>	Low
STU-020	Check duplicate — existing match	POST <code>/api/students/check-duplicate</code> with matching names (<code>First_Name: "አማኑኤል", Father_Name: "ተክሌ", Grandfather_Name: "ገብረማርያም", Mothers_Name: "ወለተ ዮሐንስ", Sex: "Male"</code>)	200 OK <code>{ exists: true }</code>	High
STU-021	Check duplicate — no match	POST <code>/api/students/check-duplicate</code> with unique name	200 OK <code>{ exists: false }</code>	High
STU-022	Check duplicate — missing fields	POST <code>/api/students/check-duplicate</code> omitting <code>Mothers_Name</code>	400 <code>Missing required fields</code>	Medium
STU-023	Count students by grade & year	POST <code>/api/students/count</code> <code>{ "academicYear": "2018", "grade": "አንደኛ ክፍል", "classification": "Regular" }</code>	200 OK <code>{ count, academicYear, grade, classification }</code>	High
STU-024	Count students — missing params	POST <code>/api/students/count</code> without <code>academicYear</code>	400 <code>Academic year and grade are required</code>	Medium
STU-025	Total students count	GET <code>/api/students/total</code>	200 OK <code>{ total: 1250 }</code>	Low

ID	Test Case	Steps	Expected Result	Priority
STU-026	Delete student	DELETE <code>/api/students</code> with <code>{ "id": "{studentId}" }</code>	200 OK <code>{ message: "Student deleted" }</code>	High
STU-027	Delete student — invalid ID	DELETE <code>/api/students</code> with invalid ID	400 <code>Valid ID is required</code>	Medium
STU-028	Classification grade options validation	Verify classification dropdowns (Regular, Extension, SignLanguage, Summer, begena)	Corresponding canonical grade options loaded	High
STU-029	Student details profile view	Open student profile in UI	Personal details, attendance history, academic results, payments visible	High
STU-030	Registration period closed	Set past <code>registrationClosedDate</code> for <code>Regular</code> 2018 in <code>/api/category-periods</code> , then attempt to POST student	403 Forbidden "Registration is closed for this category..."	High
STU-031	Registration period open	Ensure future <code>registrationClosedDate</code> , POST student	201 Created	Medium

7. Attendance

ID	Test Case	Steps	Expected Result	Priority
ATT-001	List attendance by date	GET <code>/api/attendance?date=2018-05-15</code>	Records for date returned, newest first	High
ATT-002	Filter attendance by grade & year	GET <code>/api/attendance?grade=አንደኛ ክፍል&academicYear=2018</code>	Scoped only to students in specified grade	High
ATT-003	Attendance summary mode	GET <code>/api/attendance?date=2018-05-15&grade=አንደኛ ክፍል&summary=true</code>	200 OK <code>{ total: 45, present: 42 }</code>	High
ATT-004	Summary for empty grade	GET <code>/api/attendance?grade=በገና&summary=true</code> with no students	200 OK <code>{ total: 0, present: 0 }</code>	Medium
ATT-005	Attendance pagination limit	GET <code>/api/attendance?limit=10</code>	Returns at most 10 records	Low

ID	Test Case	Steps	Expected Result	Priority
ATT-006	Mark attendance present (bulk)	POST <code>/api/attendance</code> with <code>{ date: "2018-05-15", attendance: [{ studentId, present: true, hasPermission: false }] }</code>	200 OK <code>{ success: true, insertedCount, updatedCount }</code>	High
ATT-007	Mark absent with permission	POST <code>/api/attendance</code> with <code>present: false, hasPermission: true, reason: "በህመም ምክንያት"</code>	Stored correctly with permission flag & reason	High
ATT-008	Mark absent without permission	POST <code>/api/attendance</code> with <code>present: false, hasPermission: false, reason: ""</code>	Stored correctly as unexcused absence	High
ATT-009	Idempotent bulk upsert	Submit attendance for same student and date twice	Second submit updates existing record; <code>updatedCount</code> increments	High
ATT-010	Invalid attendance payload	POST <code>/api/attendance</code> without <code>date</code> or non-array <code>attendance</code>	400 <code>Invalid request data</code>	High
ATT-011	Concurrency lock on attendance date	Two facilitators submit attendance for same date simultaneously	Distributed lock <code>lock:attendance:{date}</code> serializes writes	Medium
ATT-012	Student attendance history	GET <code>/api/attendance/{studentId}</code> with valid student ObjectId	200 OK array of historical attendance records	Medium
ATT-013	Attendance history invalid ID	GET <code>/api/attendance/invalid-id</code>	400 <code>Invalid Student ID</code>	Medium
ATT-014	QR code scanning attendance	Scan valid signed student QR code in facilitator UI	QR verified and student marked present in roster	High
ATT-015	Ethiopian calendar date mode	Test attendance with Ethiopian date picker in <code>sundays_only</code> and <code>all_days</code> modes	Dates formatted accurately in Ethiopian calendar	Medium
ATT-016	Query draft / temp attendance	GET <code>/api/attendance/temp?date=2018-05-15&markedBy=Attendance%20Facilitator</code>	200 OK draft attendance records	Low

8. Facilitators & Education Facilitators

ID	Test Case	Steps	Expected Result	Priority
FAC-001	List facilitators by role scope	GET <code>/api/facilitators</code> as HR Admin	200 OK with only Attendance Facilitator records (passwords excluded)	High
FAC-002	Get facilitator by ID	GET <code>/api/facilitators?id={facilitatorId}</code>	200 OK single facilitator document	High
FAC-003	Create Attendance Facilitator	POST <code>/api/facilitators</code> with <code>name: "ዲ/ን ቴዎድሮስ ካሳሁን", email: "facilitator.tewodros@birhanehiwot.org", password: "Facilitator123!", role: "Attendance Facilitator", grade: ["አንደኛ ክፍል"]</code>	201 Created with inserted <code>_id</code>	High
FAC-004	Create Attendance Facilitator without grade	POST <code>/api/facilitators</code> with empty <code>grade: []</code>	400 <code>Grade is required</code>	High
FAC-005	Create facilitator duplicate email	POST <code>/api/facilitators</code> with existing email	409 <code>Email already exists</code>	High
FAC-006	Update facilitator (PUT)	PUT <code>/api/facilitators</code> with updated name and grade assignment	200 <code>Updated successfully</code>	High
FAC-007	Delete facilitator	DELETE <code>/api/facilitators</code> with valid ID in requester's scope	200 <code>Deleted successfully</code>	High
FAC-008	Cross-role creation forbidden	HR Admin attempts to create Education Facilitator	403 <code>No permission for this role</code>	High
FAC-009	Toggle <code>canAddStudent</code> permission	PATCH <code>/api/facilitators/{id}</code> with <code>{ "canAddStudent": true }</code>	200 OK, <code>canAddStudent</code> updated	High
FAC-010	Total facilitators count	GET <code>/api/facilitators/total</code>	200 OK <code>{ total: count }</code>	Low

ID	Test Case	Steps	Expected Result	Priority
FAC-011	List facilitator role constants	GET <code>/api/facilitators/roles</code>	200 OK [{ value: "Attendance Facilitator", ... }, { value: "Education Facilitator", ... }]	Medium
FAC-012	Education Facilitator CRUD	Education Admin creates መርኅታ ዘርዓይ በርሄ via <code>/api/education-facilitators</code>	201 Created; full CRUD operational	High
FAC-013	Education Facilitator unauthorized access	Attendance Facilitator calls <code>/api/education-facilitators</code>	403 Forbidden	High
FAC-014	Password exclusion audit	Inspect GET <code>/api/facilitators</code> and <code>/api/education-facilitators</code> responses	Password field is never returned	High
FAC-015	Facilitator edit form UI	Update grades in HR Facilitators management UI	Updated assigned grades reflected in facilitator view	High

9. Teachers, Subjects & Assignments

ID	Test Case	Steps	Expected Result	Priority
TCH-001	List teachers	GET <code>/api/teachers</code> as Education Admin	200 OK array of teachers (passwords excluded)	High
TCH-002	Create teacher	POST <code>/api/teachers</code> with name: "መምህር አስፋው ወልደሥላሴ", email: "teacher.asfaw@birhanehiwot.org", grade: "አንደኛ ክፍል", assignedSubjects: ["ስርዓተ ቤተክርስቲያን", "ነገረ ሃይማኖት"]	201 Created (password defaults to Teacher123! if omitted)	High

ID	Test Case	Steps	Expected Result	Priority
TCH-003	Create teacher duplicate email	POST <code>/api/teachers</code> with existing email	409 Email already exists	High
TCH-004	Update teacher (PUT)	PUT <code>/api/teachers</code> with updated assigned subjects	200 Updated successfully	Medium
TCH-005	Delete teacher	DELETE <code>/api/teachers</code> with valid ID	200 Deleted successfully	High
TCH-006	Teacher forbidden for unauthorized roles	HR Admin calls <code>/api/teachers</code>	403 Forbidden	High
SUB-001	List grouped subjects	GET <code>/api/subjects?academicYear=2018&grade=አንደኛ ክፍል</code>	200 OK grouped doc { academicYear, grade, gradeNumber, subjects[] }	High
SUB-002	Add subject to grade group	POST <code>/api/subjects</code> { "name": "ስርዓተ ቤተክርስቲያን", "grade": "አንደኛ ክፍል", "gradeNumber": 1, "academicYear": "2018" }	201 Created / Added to group	High
SUB-003	Duplicate subject in grade	POST same subject name for same grade & year	409 Subject already exists for this grade and academic year	Medium
SUB-004	Batch create subjects	POST <code>/api/subjects/batch</code> with 6 Orthodox Sunday School subjects (ነገረ ሃይማኖት, የመጽሐፍ ቅዱስ ጥናት, ክርስቲያናዊ ስነ ምግባር, የቤተክርስቲያን ታሪክ, ግዕዝ ቋንቋ, መዝሙር)	200 OK { results, summary: { total: 6, created: 6, exists: 0, errors: 0 } }	Medium

ID	Test Case	Steps	Expected Result	Priority
SUB-005	Batch create empty array	POST <code>/api/subjects/batch</code> with <code>{ "subjects": [] }</code>	400 Subjects array is required and must not be empty	Medium
SUB-006	Remove subject from group	DELETE <code>/api/subjects/{groupId}</code> with <code>{ "name": "ስርዓተ ቤተክርስቲያን" }</code>	200 OK; subject removed from array; group deleted if empty	Medium
SUB-007	Remove subject missing name	DELETE <code>/api/subjects/{groupId}</code> with <code>{}</code>	400 Subject name is required in request body	Low
TAS-001	List teacher assignments	GET <code>/api/teacher-assignments</code>	200 OK array of assignments with resolved teacher and subject names	Medium
TAS-002	Create teacher assignment	POST <code>/api/teacher-assignments</code> with <code>{ teacherId, subjectId, grade: "አንደኛ ክፍል", academicYear: "2018" }</code>	201 Created	Medium
TAS-003	Duplicate teacher assignment	POST same teacher, subject, grade, and year	409 Assignment already exists...	Medium
TAS-004	Delete teacher assignment	DELETE <code>/api/teacher-assignments/{id}</code>	200 Assignment deleted successfully	Medium

10. Category Periods (Registration Windows)

ID	Test Case	Steps	Expected Result	Priority
PER-001	List all category periods	GET <code>/api/category-periods</code>	200 OK array of category periods sorted by classification	High
PER-002	Filter periods by academic year	GET <code>/api/category-periods?academicYear=2018</code>	Returns only periods matching year 2018	Medium

ID	Test Case	Steps	Expected Result	Priority
PER-003	Create or upsert category period	POST <code>/api/category-periods</code> with { <code>classification: "Regular", academicYear: "2018", startDate: "2018-01-01", endDate: "2018-12-30", registrationClosedDate: "", isActive: true }</code>	200 OK upserted period document	High
PER-004	Missing classification or year	POST <code>/api/category-periods</code> omitting <code>classification</code>	400 <code>classification and academicYear are required</code>	High
PER-005	Closed registration blocks student creation	Set <code>registrationClosedDate</code> to a past date; attempt to POST student in that classification & year	403 Forbidden "Registration is closed for this category..."	High

11. Student Results

ID	Test Case	Steps	Expected Result	Priority
RES-001	List all student results	GET <code>/api/student-results</code>	200 OK array of all student exam results	High
RES-002	Filter results by student	GET <code>/api/student-results?studentId={studentId}</code>	Returns results for specified student only	High
RES-003	Create student result	POST <code>/api/student-results</code> with { <code>studentId, studentName: "አማኑኤል ተክሌ ገብረማርያም", subjectId, subjectName: "ስርዓተ ቤተክርስቲያን", academicYear: "2018", assignment1: 9, assignment2: 10, midTest: 28, finalExam: 48, remarks: "እጅግ በጣም ጥሩ ውጤት" }</code>	201 Created; server calculates <code>totalScore: 95, average: 95, grade: "A+", and recordedDate</code>	High
RES-004	Missing required result fields	POST <code>/api/student-results</code> without <code>subjectId</code>	400 Missing required fields: <code>studentId, subjectId, academicYear</code>	High

ID	Test Case	Steps	Expected Result	Priority
RES-005	Duplicate student/subject/year result	POST result for student and subject that already has scores in 2018	409 Result already exists for this student/subject/year	High
RES-006	Letter grade threshold A+ (≥ 90)	Total score = 90	Letter grade = "A+"	Medium
RES-007	Letter grade threshold B (70–74)	Total score = 72	Letter grade = "B"	Medium
RES-008	Letter grade threshold D (45–49)	Total score = 47	Letter grade = "D"	Medium
RES-009	Letter grade threshold F (< 45)	Total score = 40	Letter grade = "F"	Medium
RES-010	Get student result by ID	GET <code>/api/student-results/{id}</code>	200 OK result document, or 404 if not found	Medium
RES-011	Update result (PUT)	PUT <code>/api/student-results/{id}</code> with new scores	200 OK; <code>totalScore</code> and letter <code>grade</code> recalculated automatically	High
RES-012	Delete student result	DELETE <code>/api/student-results/{id}</code>	200 Result deleted successfully	Medium
RES-013	UI — Student results grade report	Open student profile → Results tab	All enrolled subject scores and computed grades displayed	High

12. Payments (13 Ethiopian Months)

ID	Test Case	Steps	Expected Result	Priority
PAY-001	Lazy initialization for new student	GET <code>/api/payment?year=2018&studentId={studentId}</code> for a newly created student	200 OK; all 13 Ethiopian months (Meskerem through Pagumē) initialized to "Not Paid"	High
PAY-002	Missing payment query params	GET <code>/api/payment</code> without <code>year</code> or <code>studentId</code>	400 Year and studentId are required	High

ID	Test Case	Steps	Expected Result	Priority
PAY-003	Save monthly payment status	POST <code>/api/payment</code> with { <code>year: "2018"</code> , <code>studentId: "{studentId}"</code> , <code>data: { "Meskerem": { "status": "Paid", "amount": "500" }, "Tikimt": { "status": "Paid", "amount": "500" } }</code> }	200 OK <i>Saved successfully</i>	High
PAY-004	Partial payload preservation	POST update with only <i>Hidar</i> paid	Other months remain unaffected (normalized against <i>ETHIOPIAN_MONTHS</i>)	Medium
PAY-005	Invalid payment body	POST <code>/api/payment</code> with non-object <i>data</i>	400 <i>Invalid request</i>	Medium
PAY-006	UI payment toggle	In Student profile → Payment tab, toggle <i>Tahsas</i> to Paid with amount <i>500 ETB</i>	Payment status saved and persists across reloads	High
PAY-007	UI payment revert	Revert <i>Tahsas</i> to Not Paid	Status updated and reflected in UI	Medium
PAY-008	Legacy payment lookup route	GET <code>/api/students/{studentId}/payments</code>	200 OK array of payment records	Medium

13. Student Requests Approval Workflow

ID	Test Case	Steps	Expected Result	Priority
REQ-001	Submit student registration request	Facilitator POSTs to <code>/api/student-requests</code> with full <i>studentData</i> , <i>requestedBy: "Attendance Facilitator"</i> , <i>requestedByName: "ዲ/ን ቴዎድሮስ ካሳሁን"</i>	201 Created with { <code>_id</code> }; status saved as <i>pending</i>	High
REQ-002	Submit request missing <i>studentData</i>	POST <code>/api/student-requests</code> without <i>studentData</i>	400 <i>studentData is required</i>	High
REQ-003	List pending student requests	GET <code>/api/student-requests?status=pending</code>	200 OK array of pending requests	High

ID	Test Case	Steps	Expected Result	Priority
REQ-004	Get student request by ID	GET <code>/api/student-requests/{id}</code>	200 OK full request document	High
REQ-005	Approve student request	Super Admin PATCHes <code>/api/student-requests</code> with <code>{ id, status: "approved", approvedBy: "Super Admin" }</code>	200 Request updated; processedAt timestamp set	High
REQ-006	Reject student request with reason	Super Admin PATCHes with <code>{ id, status: "rejected", approvedBy: "Super Admin", rejectionReason: "መረጃው ያልተሟላ ነው" }</code>	200 Request updated; rejection reason stored	High
REQ-007	Patch invalid request ID	PATCH <code>/api/student-requests</code> with invalid ID	400 Valid ID is required	Medium
REQ-008	Patch unknown request ID	PATCH <code>/api/student-requests</code> with non-existent ObjectId	404 Request not found	Medium
REQ-009	UI — Facilitator request submission	Fill new student request modal as facilitator	Request appears in Super Admin pending approval list	High
REQ-010	UI — Admin request approval	Super Admin approves request in dashboard	Student record created in active roster	High

14. QR Code Generation & Verification

ID	Test Case	Steps	Expected Result	Priority
QR-001	QR generated on student creation	Create student when <code>QR_SECRET</code> is configured	Signed <code>qr_code</code> data URL generated and stored	High
QR-002	Graceful fallback when <code>QR_SECRET</code> missing	Create student without <code>QR_SECRET</code>	Student created successfully (201); <code>qr_code</code> omitted without failing creation	Medium
QR-003	Verify authentic signed QR text	POST <code>/api/qr/verify</code> with <code>{ text: "<signed_qr_text>" }</code>	200 OK <code>{ uniqueId: "BH-2018-001" }</code>	High
QR-004	Verify tampered QR text	POST <code>/api/qr/verify</code> with modified QR payload	400 Bad Request Invalid QR code with reason code	High

ID	Test Case	Steps	Expected Result	Priority
QR-005	Verify empty QR text	POST <code>/api/qr/verify</code> with <code>{ text: "" }</code>	400 <code>text is required</code>	Medium
QR-006	Stream student QR image PNG	GET <code>/api/students/{id}/qr</code>	200 OK with <code>Content-Type: image/png</code> binary stream	High

15. Reports & Export

ID	Test Case	Steps	Expected Result	Priority
RPT-001	Admin general reports page	Super Admin navigates to <code>/admin/reports</code>	Summary cards, attendance statistics, and grade charts render	Medium
RPT-002	HR reports page	HR Admin navigates to <code>/hr/reports</code>	Attendance rate and facilitator logs visible	Medium
RPT-003	Education reports page	Education Admin navigates to <code>/education/reports</code>	Academic scores, grade distribution, and teacher assignments visible	Medium
RPT-004	Facilitator scoped reports	Attendance Facilitator navigates to reports	Only assigned grade attendance figures displayed	Medium
RPT-005	Export Excel attendance report	Click Export Excel on attendance report table	Downloads valid <code>.xlsx</code> spreadsheet matching Ethiopian date range	Medium
RPT-006	Export PDF student results report	Click Export PDF on results / attendance table	Downloads formatted PDF document	Medium
RPT-007	Filter reports by Ethiopian date range	Select start & end dates in Ethiopian calendar picker	Data filtered precisely to selected period	Medium
RPT-008	Filter reports by classification & grade	Select <code>Regular</code> and <code>አንደኛ ክፍል</code>	Reports scoped strictly to chosen class and grade	Medium

16. Negative & Security Edge Cases

ID	Test Case	Steps	Expected Result	Priority
NEG-001	Malformed MongoDB ObjectId	Pass <code>12345</code> to any <code>[id]</code> route	400 Bad Request with descriptive invalid ID message	Medium
NEG-002	Non-existent resource ID	Pass valid 24-char ObjectId not in database	404 Not Found error	Medium

ID	Test Case	Steps	Expected Result	Priority
NEG-003	NoSQL injection in login/queries	Submit {"\$gt": ""} in email or parameters	Sanitized safely via regex escaping; auth fails securely	High
NEG-004	XSS injection in Amharic student name	Submit <script>alert('xss')</script> in First_Name	Stored safely; escaped properly in React UI rendering	High
NEG-005	Oversized base64 photo upload	Submit image > 5MB	Handled gracefully with client/server validation	Low
NEG-006	Distributed lock degradation	Redis server offline	Concurrency wrapper gracefully falls back; operations still complete	Medium
NEG-007	Missing Content-Type header	POST JSON body without Content-Type: application/json	Handled gracefully with 400 Bad Request	Low
NEG-008	Role tampering in request body	Send userRole: "Super Admin" as an unauthenticated or facilitator user	Role-checked endpoints verify NextAuth JWT server-side, ignoring body claim	High

17. Test Execution Checklist

Prerequisites

- ☐ MongoDB connection verified (MONGODB_URI active)
- ☐ NEXTAUTH_SECRET and optional QR_SECRET configured in .env.local
- ☐ Next.js server running (npm run dev or production URL)
- ☐ Postman collection & environment imported
- ☐ Super Admin initialized via POST /api/setup/super-admin

Smoke Test (15 minutes)

- ☐ **AUTH-012 & AUTH-001:** Setup and log in as Super Admin (superadmin@birhanehiwot.org)
- ☐ **MOB-001:** Verify mobile app login (POST /api/mobile/auth)
- ☐ **STU-006:** Register a student (BH-2018-001 - አማኑኤል ተክሌ)
- ☐ **ATT-006:** Mark attendance for the student
- ☐ **ATT-003:** Verify attendance summary returns accurate count
- ☐ **FAC-003:** Create Attendance Facilitator (ዱ/ን ቴዎድሮስ ካሳሁን)
- ☐ **SUB-002:** Add subject ስርዓተ ቤተክርስቲያን for አንደኛ ክፍል
- ☐ **RES-003:** Record exam score and verify university letter grade A+
- ☐ **PAY-003:** Mark Meskerem payment status as Paid (500 ETB)
- ☐ **AUTH-008:** Log out successfully

18. API Endpoint Reference Table

Method	Endpoint	Description	Auth / RBAC
GET	/api/auth/csrf	Get CSRF token	Public
POST	/api/auth/callback/credentials	Web credentials login	Public
GET	/api/auth/session	Get active session	Session Cookie
POST	/api/auth/signout	Web logout	Public
POST	/api/mobile/auth	Mobile JWT login	Public
OPTIONS	/api/mobile/auth	Mobile CORS preflight	Public
POST	/api/setup/super-admin	Bootstrap Super Admin	Public (one-time)
GET, POST, DELETE	/api/students	List, create, & delete students	Session Cookie / Role rules
GET, PUT, PATCH	/api/students/[studentId]	Profile, photo & QR update	Session Cookie
POST	/api/students/check-duplicate	Name+Sex duplicate check	Session Cookie
POST	/api/students/count	Count by grade & year	Session Cookie
GET	/api/students/total	Total student count	Session Cookie
GET	/api/students/[studentId]/qr	Stream student QR PNG image	Public / Session Cookie
GET	/api/students/[studentId]/payments	Legacy payments history	Session Cookie
GET, POST	/api/attendance	List attendance & bulk upsert	Session Cookie
GET	/api/attendance/[studentId]	Student attendance history	Session Cookie
GET	/api/attendance/temp	Temporary / draft attendance	Session Cookie
GET	/api/facilitators/roles	Facilitator roles list	Public / Session Cookie

Method	Endpoint	Description	Auth / RBAC
GET, POST, PUT, DELETE	/api/facilitators	Manage facilitators	Super Admin / HR Admin
PATCH	/api/facilitators/[id]	Toggle <code>canAddStudent</code>	HR Admin / Super Admin
GET	/api/facilitators/total	Total facilitators count	Session Cookie
GET, POST, PUT, DELETE	/api/education-facilitators	Education facilitators	Education Admin / Super Admin
GET, POST, PUT, DELETE	/api/teachers	Teacher profile management	Education Admin / Super Admin
GET, POST	/api/subjects	Grouped subjects list & add	Session Cookie
POST	/api/subjects/batch	Batch create subjects	Session Cookie
DELETE	/api/subjects/[id]	Remove subject from group	Session Cookie
GET, POST	/api/teacher-assignments	Teacher-subject mappings	Session Cookie
DELETE	/api/teacher-assignments/[id]	Remove assignment	Session Cookie
GET, POST	/api/category-periods	Registration windows	Super Admin
GET, POST	/api/student-results	Student exam results	Education Admin / Facilitator
GET, PUT, DELETE	/api/student-results/[id]	Result document by ID	Education Admin / Facilitator
GET, POST, PATCH	/api/student-requests	Student request workflow	Facilitators & Super Admin
GET	/api/student-requests/[id]	Get student request by ID	Session Cookie
GET, POST	/api/payment	13-Month Ethiopian payments	Session Cookie
GET, POST, PUT, DELETE	/api/admin-users	Department admins	Super Admin only

Method	Endpoint	Description	Auth / RBAC
POST	/api/qr/verify	HMAC QR code verification	Public / Session Cookie
GET	/api/sheet/health	Google Sheets health check	Session Cookie
GET	/api/sheet	Fetch Google Sheets rows	Session Cookie
GET	/api/cron/aggregate-attendance	Aggregate attendance cron	Internal / Admin